

Quantum Mechanics III

Homework 2

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Topic

Generalized Measurements and Instruments

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Generalized Measurements and Instruments. State all assumptions and conventions.

$$E_k \geq 0, \sum_k E_k = I$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$p_k = \text{Tr}[\mathcal{M}_k(\rho)]$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$\rho_k = \mathcal{M}_k(\rho)/p_k$$

Show enough intermediate work that another student can reproduce the calculation.